R&D Practical Project

Information and organization

Academic Year 2026–2027

Learning objectives

The key objective is to gain hands-on experience by applying the techniques and algorithms covered in the lectures to problems in modelling and understanding human behavior. You are welcome to explore beyond the lecture material.

The project exposes you to a current, widely used dataset and all stages of model development: ideation, reading related literature, developing a deep learning model, and evaluating it on a computer vision task.

You will also participate in a benchmarking competition. This mimics a scientific grand challenge, in which multiple groups work and compete towards a common goal.

Project teams

The project is carried out in teams of three students (exceptionally four). Students may choose their own team partners. Teaching staff can facilitate team formation for students without a team or combine smaller teams.

All teams work on the same dataset and complete both tasks.

Timeline and deliverables

Date	Milestone and deliverables
02/10/2026	Project introduction, dataset and task introduction, and team assignment. Start researching related papers and understanding the benchmark.
13/11/2026	Mid-term presentation: a short talk on progress, results, and challenges. **Deliverable:** mid-term presentation.
11/12/2026	Final presentation and project defense; final submission deadline. **Deliverables:** final presentation, report in conference paper format, code and supplementary material used to develop the solution, and poster in PDF format.

Grading

The practical project accounts for **100% of the grade for the R&D Experience Project component**.

The project grade is a whole number between 0 and 20 and is awarded per team. Individual scores may be assigned based on the team's realizations, individual contributions, and answers during the defense or practical sessions.

Component	Weight	Evaluation criteria
Mid-term presentation	15%	Quality of the mid-term presentation.
Final presentation	30%	Quality of the final presentation and project defense.
Final report	20%	Quality of the report in conference paper format.
Quality of the solution	25%	Results, code organization, reproducibility, documentation, and implementation quality.
Poster	10%	Poster quality and ability to communicate the project clearly and efficiently.

Dataset and tasks

All teams work with **EPIC-KITCHENS-100**, a large-scale egocentric audio-visual dataset of daily kitchen activities recorded from a first-person perspective. It contains approximately 100 hours of video and fine-grained action annotations, with 97 verb classes and 300 noun classes.

Task 1 – Action Recognition (AR): assign a **(verb, noun)** label to each trimmed video segment, recognizing the action taking place. For example, segment `P01_01_0` is labelled `(3, 3)`, corresponding to *(open, door)*.

These tasks go beyond image classification: the model must recognize actions in video. Although the dataset contains both video and audio, **both tasks use the video modality only**.

Task 2 – Short-Term Action Anticipation (STA): predict the upcoming action's verb and noun using video observed up to **one second before the target action starts**. For an event starting at 20 seconds, the latest permitted observation is at 19 seconds. Observation length is a model-design choice; five seconds of context from 14 to 19 seconds is a possible baseline. Do not use frames in the one-second gap or the target action, including during feature extraction.

Each team must complete **both AR and STA**. AR is scored by Top-1 and Top-5 accuracy for verbs, nouns, and actions, with Action Top-1 as its primary metric. STA is scored by **class-mean Top-5 recall (MT5R)**, with Action MT5R as its primary metric. MT5R averages per-class Top-5 recall equally over classes present in the evaluated subset. Both tasks also report tail-class and unseen-participant results.

Training, validation, and hidden test data

Use the **provided training annotations only** for development. Each team chooses and documents its own train/validation split within the provided train set.

The **provided test split is the hidden test set**, you will receive `EPIC_100_test.csv`, a custom event table with IDs, timestamps, frame boundaries, and `sta_observation_end_seconds` (start time minus 1.0), without narrations or semantic labels. Do not retrieve or use the public official

labels for training, tuning, or local evaluation.

All hidden-test events are retained for both tasks. For actions starting before one second, no earlier video is available: use an empty or padded observation and a prior-based STA prediction without reading future frames.

Data access and competition submissions

The downscaled videos (`640x360`, approximately 17 GB) and annotations are provided through KU Leuven ManGO at `/set/home/ciis-lab/datasets/epic-kitchens`. Log in with your KU Leuven account at `https://mango.kuleuven.be`. iCommands can be used to download the data to VSC compute instances.

Each student must register on `https://codabench.org/` using their official student email address. Each team must create an organization named `Group X`, where `X` is its assigned group number, and invite all team members.

Submit a ZIP archive containing `submission.pt` (AR) and `submission_sta.pt` (STA) at its root, both saved with `torch.save`, and select both Codabench tasks. Each file must contain one prediction dictionary per hidden-test event, with keys `narration_id`, `verb_output`, and `noun_output`. The outputs must be floating-point score vectors of lengths 97 and 300 respectively (logits or probabilities), rather than predicted class indices. Joint action scores are derived from the verb and noun vectors. Missing, duplicate, or unexpected event IDs and non-finite scores are rejected.

The competition allows five submissions per day and 50 in total. Consult the accompanying dataset-access instructions and competition guide for the full submission procedure and detailed feedback.

Important information

Using the same assignment or project in multiple courses is not allowed. If there is overlap with another course where you may choose the dataset, or with your master's thesis, contact the teaching staff early.

Use of generative AI

Generative AI may be used as a tool in your workflow. Follow KU Leuven's guidelines for responsible use and cite or credit its use appropriately in your work: `https://www.kuleuven.be/english/genai`.